



Retake solution Cognitive System

Kognitive Systeme (Technische Universität München)



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Esolution

Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Cognitive Systeme: Neural Networks and Applied AI

Exam: IN2222 / Retake

Date: Thursday 5th October, 2023

Examiner: Prof. Knoll

Time: 11:00 – 12:15

	P 1	P 2	P 3	P 4	P 5	P 6
I						

Working instructions

- This exam consists of **20 pages** with a total of **6 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 150 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **analog dictionary** English ↔ native language
- Give **concise** answers using appropriate **terminology**.
- You may only answer in **English**.
- **Do not** write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- You have **75 minutes** working time

Left room from _____ to _____ / Early submission at _____

Problem 1 Cognition and Cognitive Science (24 credits)

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a) Provide a brief and concise definition of the term cognitive system.

A cognitive system is an autonomous system (1) that can perceive (1) its environment, learn (1) from experience, anticipate the outcome of events, act (1) to pursue goals, and adapt to changing circumstances.

C02-1 S17

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b) List and define the three types of cognitive capabilities mentioned in the lecture. For each type provide at least two specific subcapabilities.

- Self-Reliance: Goal-directedness, Autonomy, Interaction
- Perception and Action: Interpretation, Sensing, Anticipation, Action
- Adaption: Reaction, Learning, Anomaly Detection

C02-1 S22-S24

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c) Neuroimaging is the use of quantitative (computational) techniques to study the structure and function of the central nervous system, developed as an objective way of scientifically studying the healthy human brain in a non-invasive manner.

State the full names of the two nuclear magnetic resonance based methods that are introduced in the lecture.

Magnetic Resonance Imaging (MRI) and Diffusion Tensor Imaging (DTI).

C03-1 S10

d) List and explain two main properties of dynamical systems.

0
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Dissipation: the number of reachable states reduces over time
Non-Equilibrium System: stable function requires an external "energy" supply
Non-Linearity: complex behavior can emerge from a small set of state parameters
Collective Variables: the system is represented by a small set of state variables
1 point for every correct name and 1 point for every correct explanation
C03-2 S 38

e) Provide a brief description of the main characteristics of enactive emergent systems and explain the two principles that they are based on.

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An enactive emergent system develops its own understanding of the world around it (1) through its interactions with the environment (1).
Enaction is governed by two key mechanisms:

- Phylogeny (1): The interaction between the system and its environment is structurally determined (1) by the innate embodied physical and cognitive capabilities (1).
- Ontogeny (1): The system is structurally coupled with the environment through sense-making (1) and generates its own specific epistemology by capturing the regularities of interaction (1).

C03-2 S 37

Problem 2 Cognitive Architectures and Neurorobotics (23 credits)

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4	☐

a) Answer the following questions about ACT-R, which stands for adaptive control of thought-rational.

- State the cognitive architecture type that ACT-R belongs to.
- State what ACT-R tries to implement.
- Name the programming language that ACT-R is written in.
- Name several of the ACT-R basis components and modules.

Adaptive control of thought-rational (ACT-R) is a hybrid cognitive architecture (1) and implements the principles of a unified theory of cognition (1). ACT-R is a software framework written in Common Lisp (1).

- ACT-R is based on modules, buffers, and a production system (1)
- ACT-R has four main modules for intention, declarative memory, vision, and action and a procedural module (1)

(one point per item, max four points)

E09 S24

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b) In ACT-R, a production is a statement of a particular contingency that controls behavior. Every production is identified by a name and is assembled from a list of buffer operations.

Write the syntax to define a general production as taught in the lecture.

```
(p ProductionName
  buffer tests (1)
==> (1)
  buffer changes and requests (1)
)
```

E09 S32

0	☐
1	☐
2	☐
3	☐
4	☐

c) Name four advantages of executing experiments with virtual robots instead of physical ones.

- Increased simulation speed
- Change parameters to simplify experiments
- No sensor imperfections
- Repeat runs without deviations
- Reset experiment automatically
- No need for expensive robots
- No risk of breaking robots

1 point for every correct advantage, 4 points in total

E03, S5

d) Describe the buffer test and buffer action of the following production rule using ACT-R terminology.

```
(p type
  =goal>
    isa goal
    state enter-number
==>
  +MANUAL>
    CMD PRESS-KEY
    KEY "2"
)
```

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Test if the goal buffer (1) holds a chunk of type goal (1) that indicates the variable "state" as "entering-number" (1), then send a buffer request (1) to the manual module (1) to press the key "2" (1)

E09 S46

e) The Neurorobotics Platform (NRP) has been designed to test brain models and cognitive architectures. Name three tools that are part of the NRP and mention the purpose of each.

- Robot Designer: Robot modeling
- State Machine: Scripted environment actions during simulation
- Transfer Functions: Python interface to connect robots with brains
- (Morphological) Brain Visualizer: Visualization of populations, spikes and neuron location
- Environment Designer: Custom environment design
- Virtual Coach: A python programming interface to script simulations
- Plotting/CSV Recording: Data visualization / logging

(one point per tool and 1 point per description)

E03 S22

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Problem 3 Human Brain (18 credits)

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2 ☐

a) Describe the purpose of glial cells.

Glial cells have a supporting function and do not play a direct role in information processing. (2)
E04 S27

0 ☐
1 ☐
2 ☐

b) Name the anatomical structure in the brain that plays a key role in the formation of episodic long-term memories.

Hippocampus (2)
C04-2 S27

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3 ☐
4 ☐
5 ☐
6 ☐

c) Modern AI is considered to start with the computational model and perceptron model developed by McCulloch and Pitts, and Rosenblatt, respectively. Name three features of these models.

- Two-state neurons (fire and rest)
- Firing only after crossing a threshold (All-or-none)
- Weight adaptation
- Classification of input patterns
- Initially modelled with motors and potentiometers

2 points per correct answer, max 6 points
C04-2 S36-37

d) Name and describe the two subsystems of the vertebrate nervous system.

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Central Nervous System (CNS) (1) - It collects and distributes data throughout the body (1).
Peripheral Nervous System (PNS) (1)- The peripheral nervous system transmits signals between sensory organs, muscles and internal organs and the CNS (1).
C04-1 S6

e) Write the following neuron models in order according to their complexity. Start with the least complex one.

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1. FitzHugh-Nagumo model
2. McCulloch-Pitts model
3. Hodgkin-Huxley model
4. Leaky integrate-and-fire model
5. AdEx integrate-and-fire model

1 point for three items in the correct position, 1 extra point for four items in the correct position, 2 extra points for all items in the correct position

1. McCulloch-Pitts Model (E04, Slide 33)
2. Leaky Integrate-and-Fire (LIF) (E04, Slide 45-46)
3. AdEx Model (E04, Slide 48)
4. FitzHugh-Nagumo model (E04, Slide 44)
5. Hodgkin-Huxley (E04, Slide 40)

Problem 4 Biological Neurons and Spiking Neural Networks (29 credits)

The membrane potential of a simplified Leaky Integrate and Fire (LIF) neuron with constant input current I is given by

$$v(t) = -RI \cdot e^{-\frac{t}{RC}} + RI$$

with membrane capacity C and membrane resistance R . Spikes of LIF neurons are generated if the membrane potential reaches the threshold value v_{thresh} . After the occurrence of a spike the membrane potential is immediately set to $v_{\text{reset}} = v_0$, i.e. we have no refractory period.



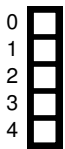
a) Determine the time of the first spike t_{spike} in the neuron model above.

time of first spike: $v(t_{\text{spike}}) = v_{\text{thresh}}$ (2 point)

- Solve equation (3 point)

$$\Rightarrow t_{\text{spike}} = -\tau \ln\left(\frac{v_{\text{thresh}} - RI}{-RI}\right)$$

Reference: Ex05 - problem sheet



b) State the condition on the input current I and the threshold v_{thresh} , such that the neuron above is spiking.

$\ln(\dots)$ has to be defined and negative, since $t_{\text{spike}} > 0$ (2 points)

$$\Rightarrow 0 < \frac{v_{\text{thresh}} - RI}{-RI} < 1 \text{ (2 points)}$$

Reference: Ex05 - problem sheet



c) Does the neuron above spike periodically, if the condition is fulfilled?

yes. (1 point)

Reference: Ex05 - problem sheet

d) State the two main schemes of encoding techniques for spiking neural networks.
Name the main advantage for each of them.

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- Frequency (en)coding OR rate (en)coding (1): Robustness against noise (1)
- Time (en)coding (1): Fast reaction time (1)
- Population coding (1): Robustness against noise (1)

EX05, slide 8

e) The Hebbian theory can be summarized as "Cells that fire together wire together".
Name the three fundamental properties for a Hebbian learning algorithm to be biologically plausible and stable over an undetermined period of time.

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Saturation (1), locality (1), or competition/selectivity (1)
Reference: EX05 - slide 11

f) Describe what means "competition" or "selectivity" in the context of learning, and what is the problem when a Hebbian learning algorithm does not have this feature. You might draw a graph for illustrating the problem.

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If a learning rule has competition or selectivity, the weights of the different synapses do not converge to the same value over large periods of time (3)

EX05 - slide 11



g) The following equation is a basic form of Hebbian learning.

$$w_t = w_{t-1} + \frac{\Delta t}{\tau_w} v \cdot u$$

Modify the equation in order to add competition to the learning rule.

A solution is to normalize the equation (2*)

$$w_t = w_{t-1} + \frac{\Delta t}{\tau_w} \left(v \cdot u - \frac{v(n \cdot u)}{n} N_u \right)$$

(7 points) * The 2 points are only added if the equation is missing
EX05 - slide 11



h) Neuroplasticity involves changes in the brain over time as an adaptation to new situations. One type of neuroplasticity is *functional plasticity*.

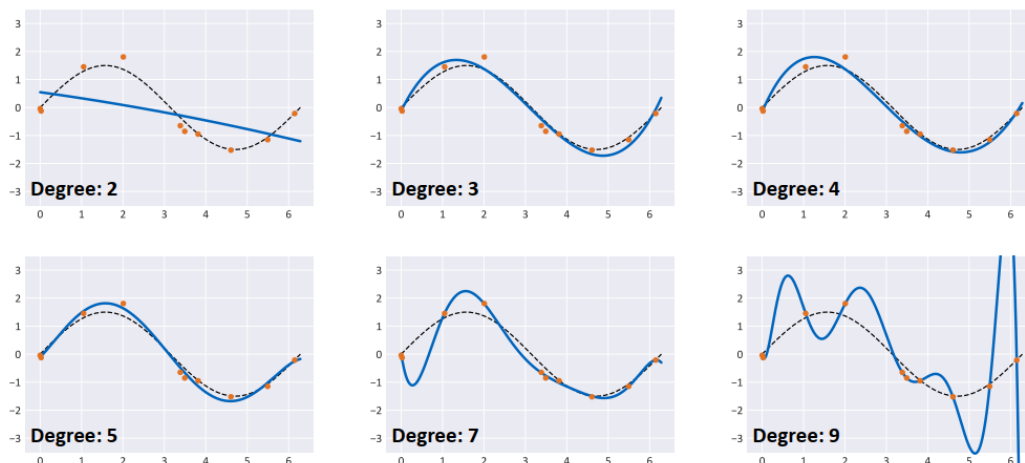
Explain the type of changes performed in the nervous system when *functional plasticity* takes place, and state if this type of neuroplasticity plays a role when it comes to learning.

functional plasticity: alter and adapt functional properties of neurons (1 points) yes, it plays a role in learning (1 point)

EX05 - part 2

Problem 5 Machine Learning (26 credits)

Your colleagues A, B and C have been working on some machine learning tasks recently. Colleague A has fitted different polynomials to a dataset which is shown below.



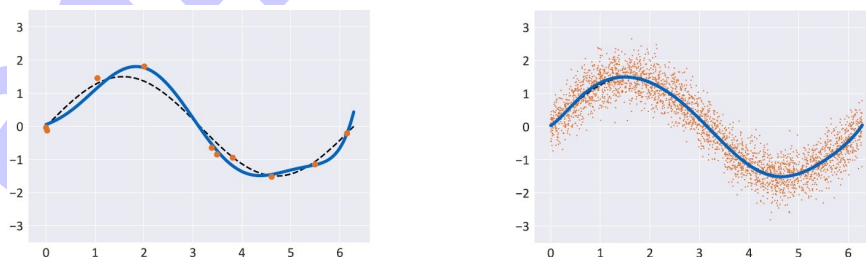
a) Which polynomial do you suggest your colleague to choose?
Explain also the reasons why the other ones should not be chosen.

- The top middle model (Degree: 3) (1).
- The top left model (Degree: 1) suffers underfitting (1),
- the bottom middle and bottom right (Degrees: 7 & 9) suffer overfitting (1).
- From the three remaining ones, it is desirable to choose a model that is as simple as possible, as stated in the principle of Occam's razor (1).

E01-2 p34-36

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b) Colleague A chose originally the model with degree 9. To avoid the issue depicted in the last question, colleague A applied two different methods and obtained the results below.



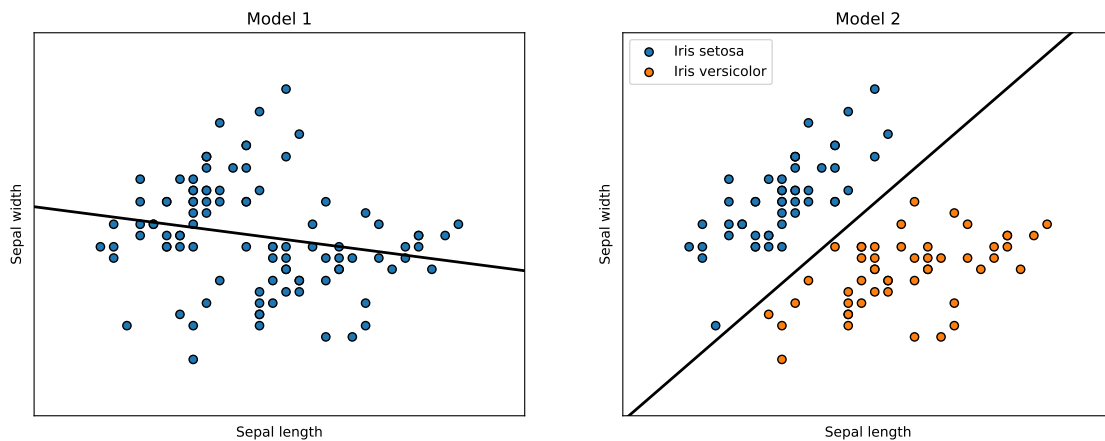
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Give the name of each method your colleague has used, and name another method that can also help avoid this issue.

- left: regularization (1).
- right: add more training data (1).
- another method: dataset augmentation/ early stopping/ use a simpler model (1).

E01-2 p42-43

Colleague B has come across the following two graphs, describing two different supervised learning models for the iris dataset. However, colleague B is unsure what the underlying models are.



c) For each of the models (Model 1 and Model 2) above, state what their input x and output y are.

- The input of Model 1 is the sepal length (1) and the output is a continuous value for the predicted sepal width (or the other way around) (1).
- The inputs of Model 2 are the sepal length and sepal width (1), and the output is the predicted iris species (or class) (1).

C05-1 p10, p11

d) For each of the models above, state the type of the task that they are performing.

- Model 1 is a regression model (linear regression) (1)
- Model 2 is a classification model (logistic regression) (1)

C05-1 p10, p11

Colleague C is working on large language models like ChatGPT.

e) Is GPT the same type of model as Model 1 or Model 2 above?
If not, state which type of model it is, and explain how this type of model is different from the two models above.

- No, large language models are generative models. **C015 p21** (1)
- Generative models generate new data, while discriminative models like Model 1 and Model 2 classify data. **C015 p21** Or explain by probability: generative models are based on prior or joint probability, while discriminative models are based on posterior probability. **E02 p37** (2)

f) GPT is based on a transformer neural network architecture that provides a feature representation of raw data. Explain two advantages of using deep neural networks over manually engineered features.

0
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4

4p max.

- Deep neural networks can learn suitable feature space representations automatically. (2)
- Expert knowledge that is encoded in manually engineered features is replaced by huge datasets. (2)
- Layers in deep neural networks correspond to features at different levels of abstraction. (2)

C05-2 p3

g) Name and explain three different learning paradigms apart from supervised learning.

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5
6

1p name, 1p explanation, 6p max

- Unsupervised learning (1): learning of statistical regularities in the input data (1)
- Reinforcement learning (1): learning which action to choose in a certain situation in order to maximize an external reward signal through interaction with the environment (1)
- Semi-Supervised learning (1): mixed dataset (labeled / unlabeled data) (1)

E01 p24

Problem 6 Reinforcement Learning (30 credits)

- 0 ☐
1 ☐
2 ☐
3 ☐
4 ☐
- a) The concept of interaction loop that occurs in humans and vertebrate animal species motivates the principal logic behind reinforcement learning methods.
- State the two distinctive requirements of such learning from interactions.
 - Provide two examples of reinforcement learning that occurs naturally in humans or animals.

Humans and animals learn from interaction with our environment **without examples** (1).
The learning is **goal-directed** (1).

Examples of RL:

- A new-born gazelle that learns running at 20 miles per hour within 30 minutes.
- Playing chess, backgammon, video games.
- Investing across multiple assets (portfolio optimization).

(one point per each, max two points)

C06 S 1:4

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3 ☐
4 ☐
5 ☐
- b) Fill in the blanks: _____ conditioning is the corresponding learning process for prediction, while _____ conditioning is the corresponding learning process for control.
Furthermore, answer the following questions.
- Briefly explain the concept of prediction and control in the context of reinforcement learning.
 - Out of the two forms of conditioning, state the one that can assume the stimulus response as an interpretation of a reinforcement learning policy.

Classical conditioning is the corresponding learning process for prediction (1).
Operant/Instrumental conditioning is the corresponding learning process for control (1).
C06 S 1:7 & S 2:14

Prediction denotes the evaluation of a value function according to a policy (policy evaluation) (1).
Control is the iterative process of evaluating a value function and updating the policy for the new value function accordingly (policy evaluation + policy improvement) (1).
C06 S 2:14

According to **operant/instrumental conditioning**, a policy can be also interpreted as the stimulus response (1).
C06 S 1:18

☐	0
☐	1
☐	2
☐	3
☐	4

c) A deterministic policy selects a specific action based on the current environment state. Mathematically, this can be described as $a = \pi(s)$ where a , s , and π are respectively the action, state, and policy.

- In contrast to this deterministic action selection, describe how a stochastic policy can be defined via a mathematical equation.
- Name one deterministic and one stochastic policy.

Stochastic policy formulation: $\pi(a | s) = \mathbb{P}[A_t = a | S_t = s]$ (2) **C06 S 1:16**

- Deterministic policy: Greedy (1) **C06 S 4:6**
- Stochastic policy: ϵ -Greedy (1) **C06 S 4:7**

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☐	5
☐	6
☐	7
☐	8
☐	9

d) Value functions are the central components in the value-based reinforcement learning methods taught in the lecture.

- Describe the value functions and their use in reinforcement learning.
- Name and define the two types of value functions, and provide the mathematical equations of each.

A value function is a function that quantifies value for every state (1).

By accumulating the knowledge that the agent has gained from the experienced returns G_t (1), the value function helps to find a policy π that maximizes the return G_t immediately at every time step t . (1)

1. State-value function (1): The expected return when a specific policy is followed after visiting a particular state (1).

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t | S_t = s] \quad (1)$$

2. Action-value function (1): The expected return when a specific policy is followed after choosing an action in a particular state (1).

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t | S_t = s, A_t = a] \quad (1)$$

C06 S 2:4

0 ☐ e) The python function below is part of a reinforcement learning agent and is called whenever the action-value function needs to be updated.

1 ☐

2 ☐

3 ☐

4 ☐

5 ☐

6 ☐

7 ☐

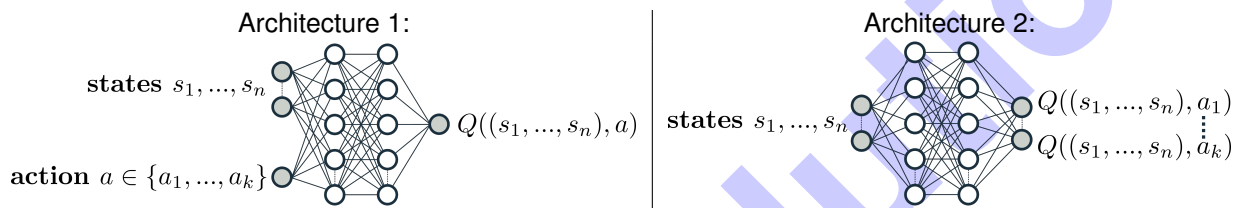
8 ☐

```
def train_model(state, action, reward, next_state, next_action):
    target = reward + gamma * model.predict(next_state)[next_action]
    values = model.predict(state)
    values[action] = target
    model.fit(state, values, epochs=1)
```

Consider the following facts for the function *train_model*.

- The object *model* is implemented as a feed-forward neural network that approximates the action-value function $q_{\pi}(s, a)$.
- The output layer of the object *model* uses a linear activation function.
- There exists k possible actions at every time step (i.e. $|A(S_t)| = k; \forall t$).

Furthermore, consider the following two illustrations of neural network architectures, which are candidates to represent the object *model*.



- State which architecture is a possible representation for the object *model* based on the context inferred from the code snippet.
- State if the implemented architecture generalizes on states only, actions only, or both.
- State how many times the implemented architecture has to be run in forward direction to find the action that maximizes the action-value function $Q(S_t, A_t)$ at every time step t .
- State the reason behind using a linear activation function in the output layer of the network representing the object *model*.

Architecture 2. (2p)

It generalizes only on states. (2p)

One time to run architecture 2 at every time step. (2p)

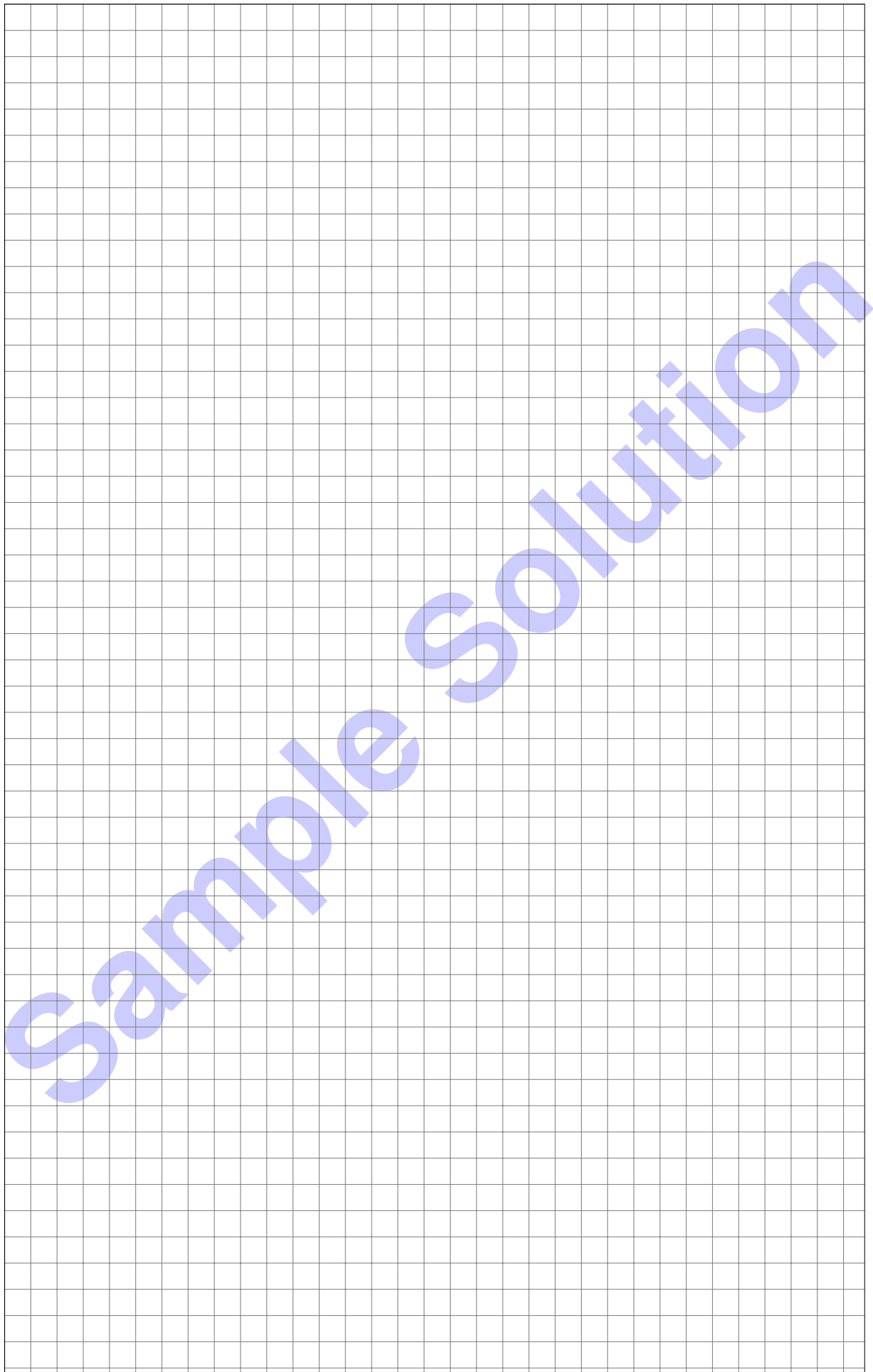
The linear activation function is used because the output layer approximates the action value for the action (1), which can take any real value (1).

E08 S 8,9,11

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Sample Solution

Sample Solution



Sample Solution